

The Cosmic Web: 0/0 at the Largest Scale

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2026

The Removable Singularity Framework

Abstract

We show that the universe itself is a 0/0 removable singularity at the LARGEST scale. The flatness problem ($\Omega = 1.000$) is the cosmic 0/0: the universe is exactly at the critical density. The cosmic web of galaxies is a scale-free network with $P(k) \sim k^{-\gamma}$ and $\gamma \sim 2.1$, the SAME as the Internet, brain, and financial networks. This is universality across ALL scales, from subatomic to cosmic. The flatness problem, dark energy, and structure formation are all connected through the 0/0 framework.

1. Flatness Problem (Cosmic 0/0)

$$\Omega = \Omega_m + \Omega_L = 1.000$$

The Friedmann equation describes the expansion of the universe. The critical density ρ_c divides recollapsing from expanding universes. $\Omega = \rho/\rho_c = 1.000$ (observed by WMAP/Planck). This is the LARGEST 0/0 in existence: the universe is exactly at the critical density.

2. Scale-Free Cosmic Web

$$P(k) \sim k^{-\gamma} \quad (\gamma \sim 2.1)$$

Galaxies form a scale-free network with the SAME structure as the Internet ($\gamma \sim 2.1$), social networks ($\gamma \sim 2.5$), and biological networks ($\gamma \sim 2.3$). Hubs are galaxy clusters (Virgo, Coma, Shapley). Filaments connect clusters. Voids are empty regions.

3. Structure Formation

Dark matter drives structure formation. Below $\Omega = 1$: universe recollapses (Big Crunch). Above $\Omega = 1$: universe expands forever. At $\Omega = 1$: flat universe (0/0). Structure formation is maximized at the critical density.

4. Dark Energy

Dark energy (cosmological constant) drives the acceleration of the universe. $\Omega_L = 0.685$, $\Omega_m = 0.315$. The age of the universe is 13.8 Gyr. The expansion is accelerating, driven by the 0/0 of dark energy.

5. Universality Across Scales

The SAME structure ($\gamma \sim 2-3$) appears at EVERY scale:

- Subatomic: quark confinement
- Atomic: electron orbitals
- Molecular: chemical bonds
- Cellular: neural networks ($\gamma \sim 2.3$)
- Social: social networks ($\gamma \sim 2.5$)
- Financial: market networks ($\gamma \sim 2.8$)
- Cosmic: galaxy web ($\gamma \sim 2.1$)

This is universality for the cosmos: same structure at ALL scales.

6. Connections

System	Prior Chapter	Connection
Scale-free networks	Ch.45 Networks	Same structure
Dark matter	Ch.20 Dark matter	Structure formation
Dark energy	Ch.21 Dark energy	Acceleration
Toomre Q	Ch.19 Toomre	Instability
Self-organization	Ch.41 SOC	Criticality
Topological defects	Ch.40 BKT	Cosmic strings
Early universe	Ch.36 Ising	Phase transitions

7. Conclusion

The cosmic web is the LARGEST 0/0 in existence. $\Omega = 1.000$ (exactly!) means the universe is flat, at the critical density. Galaxies form a scale-free network with $\gamma \sim 2.1$, the SAME as Internet, brain, and financial networks. This universality across ALL scales -- from subatomic to cosmic -- is the deepest insight of the 0/0 framework.

References

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